

EEE243

Course Title: Signals and Systems

Project Report



Inspiring Excellence

Section: 4

Group No: 6

Name of the Project: ***Designing Audio Noise Reduction System Using Frequency-Domain Techniques***

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Objective:

In practical audio communication systems, high frequency noise is usually present, which may affect the quality of voice signals. This project introduces the design and simulation of an audio noise reduction system using frequency domain approach. Both the voice signal and the noise signal are analyzed using Fourier Transform analysis to determine the spectral components of the two signals. Moreover, a 6th order Butterworth LPF is designed to differentiate the high frequency noise components from the useful voice components. The theoretical analysis is verified through MATLAB simulation based on time domain plots, frequency domain spectra, filter response analysis and Signal-to-Noise Ratio (SNR) evaluation.

The main aim of this project is to design a system that detects and removes the high-frequency noise from voice recordings and design a low-pass filter that suppresses them while preserving the original voice signal.

Problem Statement:

The recorded noisy signal is represented as:

$$y(t) = x(t) + n(t)$$

Where,

$$x(t) = \cos(2\pi 200t) + \sin(2\pi 180t) + [0.8 \sin(2\pi 250t)] e^{-2\alpha t}$$

Noise Signal:

$$n(t) = 0.5 \cos(2\pi f t) + 0.3 \sin(2\pi 6500t + \frac{2\pi}{5})$$

Here, $\alpha = 6$

Sum of all members's ID: 97184578

$f = 3870$ Hz ; High-frequency sinusoidal noise.

Theoretical Analysis:

Here, we see that the signal $x(t)$ represents the voice signal where the signal $n(t)$ is the high-frequency noise signal and the $y(t)$ signal is the combination of both the signals $x(t)$ and $n(t)$. To solve this particular problem, we began plotting the signals $x(t)$, $n(t)$ and $y(t)$ in the time domain using MATLAB. By plotting, the points help us to examine the characteristics of the signals, their mathematical expressions and identify the interference caused by noise in $y(t)$.

1. We used Fourier Transforms to decompose both $x(t)$ and $n(t)$ for enabling understanding of frequency content of the signals. This analysis helped in identifying specific frequency ranges affected by noise and those frequencies that need filtering
2. We combine the signals plotted with the mathematical calculations of Fourier Transform. By doing so, we verify both the theoretical analysis and MATLAB simulation based on time domain plots, frequency domain spectra, filter response analysis and Signal-to-Noise Ratio (SNR) evaluation.

Theoretical Frequency Domain Calculations:

The Fourier Transform converts a time-domain signal into its frequency-domain representation. Each sinusoidal component produces spectral peaks at its corresponding frequency.

Theoretical Analysis

Given signal, $y(t) = x(t) + n(t)$

where,

$$x(t) = \sin(2\pi 420t) - \cos(2\pi 190t) + [0.8 \sin(2\pi 250t)] \times e^{-12t}$$

Here, our group no. is 6

$$\alpha = 6$$
$$\therefore x(t) = \sin(2\pi 420t) - \cos(2\pi 190t) + 0.8 \sin(2\pi 250t) e^{-12t}$$

Noise signal:

$$n(t) = 0.5 \cos(2\pi f t) + 0.3 \sin(2\pi 6500t + \frac{2\pi}{5})$$

Here, $f = 3870$ Hz

[Sum of all group members id: 97184578; $Y=8$ & $Z=7$]

$$\therefore n(t) = 0.5 \cos(2\pi 3870t) + 0.3 \sin(2\pi 6500t + \frac{2\pi}{5})$$

Therefore,

$$y(t) = \sin(2\pi 420t) - \cos(2\pi 190t) + 0.8 \sin(2\pi 250t) e^{-12t} + 0.5 \cos(2\pi 3870t) + 0.3 \sin(2\pi 6500t + \frac{2\pi}{5})$$
$$\Rightarrow y(t) = \sin(840\pi t) - \cos(380\pi t) + 0.8(500\pi t) e^{-12t} + 0.5 \cos(7740\pi t) + 0.3 \sin(13000\pi t + \frac{2\pi}{5})$$

• Fourier transformation of $[\sin(840\pi t)]$:

$$\sin(\omega_0 t) \xrightarrow{F} j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

$$F\{\sin(840\pi t)\} = j\pi [\delta(\omega + 840\pi) - \delta(\omega - 840\pi)]$$

• Fourier transformation of $[-\cos(380\pi t)]$:

$$\cos(\omega_0 t) \xrightarrow{F} \pi [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

$$F\{-\cos(380\pi t)\} = -\pi [\delta(\omega + 380\pi) + \delta(\omega - 380\pi)]$$

• Fourier transformation of $[0.8 \sin(500\pi t) e^{-12t}]$:

$$\sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

$$\Rightarrow \sin(500t) = \frac{e^{j500\pi t} - e^{-j500\pi t}}{2j}$$

Now,

$$0.8 e^{-12t} \cdot u(t) \cdot \frac{e^{j500\pi t} - e^{-j500\pi t}}{2j}$$

$$= \frac{0.8}{2j} [e^{-12\pi t} \cdot e^{j500\pi t} - e^{-12\pi t} \cdot e^{-j500\pi t}] u(t)$$

$$= \frac{0.8}{2j} [e^{(-12\pi + j500\pi)t} - e^{(-12\pi - j500\pi)t}] u(t)$$

$$e^{-bt} u(t) \xrightarrow{F} \frac{1}{b + j\omega}$$

$$e^{j\omega_0 t} x(t) \longleftrightarrow x(\omega - \omega_0)$$

For $e^{(-12\pi + j500\pi)t} \cdot u(t)$:

$$\frac{1}{12\pi + j(\omega - 500\pi)}$$

For $e^{(12\pi - j500\pi)t} \cdot u(t)$:

$$\frac{1}{12\pi + j(\omega + 500\pi)}$$

$\therefore$ The full transformation is:

$$\frac{0.8}{2j} \left[\frac{1}{12\pi + j(\omega - 500\pi)} - \frac{1}{12\pi + j(\omega + 500\pi)} \right]$$

Fourier transformation of Noise component:

$$0.5 \cos(7740\pi t)$$

$$\cos(\omega_0 t) \xrightarrow{F} \pi [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

$$N_1(\omega) = 0.5\pi [\delta(\omega + 7740\pi) + \delta(\omega - 7740\pi)]$$

$$0.3 \sin(13000\pi t + \frac{2\pi}{5})$$

$$\sin(\omega_0 t + \theta) \xrightarrow{F} j\pi [e^{j\theta} \delta(\omega + \omega_0) - e^{-j\theta} \delta(\omega - \omega_0)]$$

$$N_2(\omega) = 0.3j\pi [e^{j2\pi/5} \delta(\omega + 13000\pi) - e^{-j2\pi/5} \delta(\omega - 13000\pi)]$$

∴ The total Fourier Transformation is given below:

$$\begin{aligned}
 Y(\omega) = & j\pi [\delta(\omega + 840\pi) - \delta(\omega - 840\pi)] - \pi [\delta(\omega + 380\pi) + \delta(\omega - 380\pi)] \\
 & + \frac{0.8}{2j} \left[\frac{1}{12\pi + j(\omega - 500\pi)} - \frac{1}{12\pi + j(\omega - 500\pi)} \right] \\
 & + 0.5\pi [\delta(\omega + 7740\pi) + \delta(\omega - 7740\pi)] \\
 & + 0.3j\pi [e^{j2\pi/5} \delta(\omega + 13000\pi) - e^{-j2\pi/5} \delta(\omega - 13000\pi)]
 \end{aligned}$$

Therefore, $Y(\omega) = X(\omega) + N(\omega)$. The desired components are below 420 Hz, while the explicit noise tones are at 3870 Hz and 6500 Hz.

Spectrum Analysis:

Frequency (Hz)	Signal Type	Peak Magnitude (approx)	Phase Angle
190	Voice	3.14	180°
250	Damped Voice	Broad Peak~2.5	continuos
420	Voice	3.14	-90°
3870	Noise	1.57	0°
6500	Noise	0.94	72°

Table: Spectrum Analysis

Magnitude Spectrum:

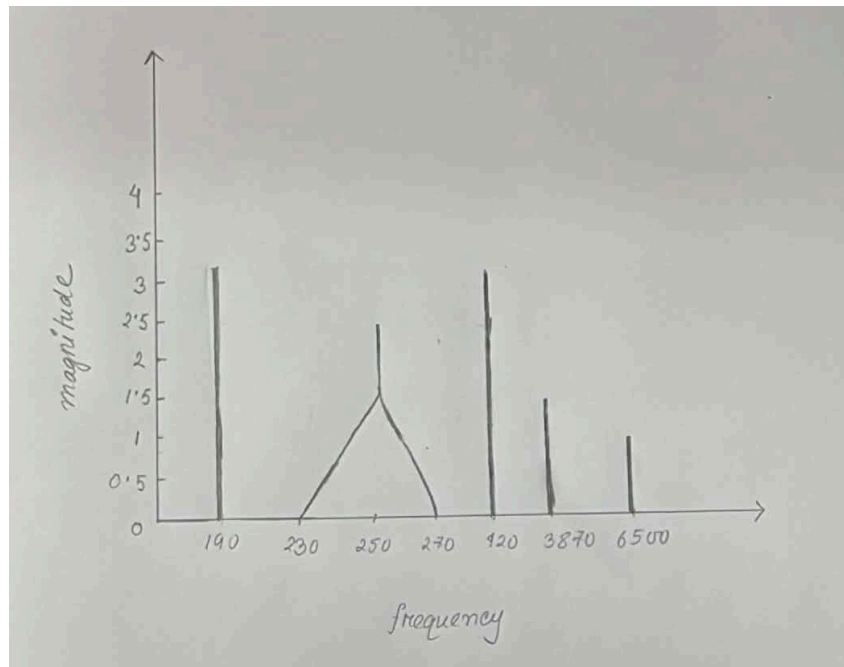


Fig: Magnitude Spectrum

Phase Spectrum:

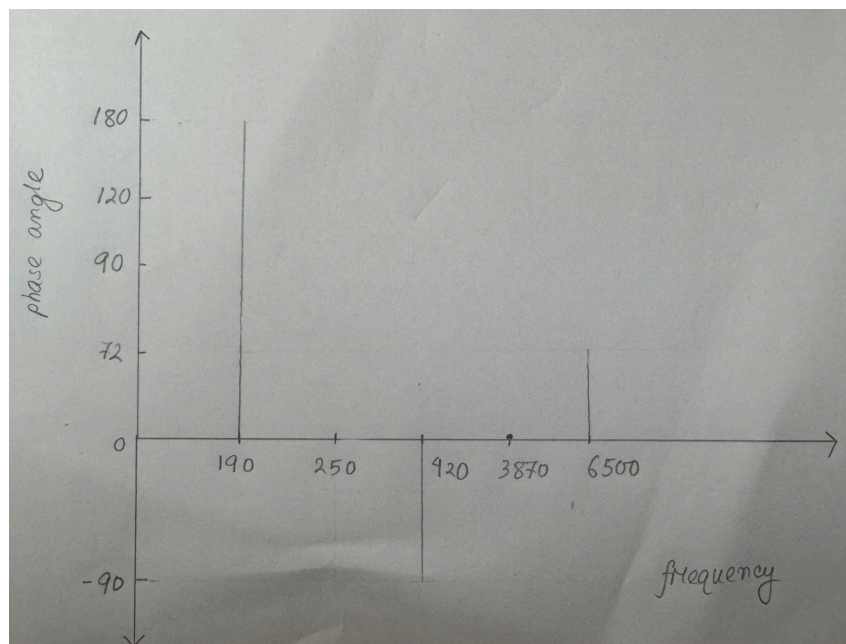


Fig: Phase Spectrum

Filter Design & Characteristics:

To attenuate the high frequency noise while keeping the original voice signal intact, we used a 6th-order Butterworth low-pass filter that is used for a smooth passband and a relatively sharp transition. The low pass filter allows low frequencies to pass through while suppressing the higher frequencies.

- Cutoff Frequency, $f_c=1000$ Hz
- Reason for Selecting 1000 Hz : All useful voice frequencies lie at 190 Hz, 250 Hz and 420 Hz
- Noise components exist at: 3870 Hz & 6500 Hz.

Simulation Analysis:

```
%noise reduction eee243

clc;

clear;

close all;

%1.Setting Up the Necessary Parameters

Fs = 50000;      % the sampling frequency in Hz

T = 1/Fs;        % sampling period from Fs

t = 0:T:0.05;    % time duration is from 0 to 0.05

alpha = 6;       % according to group number 6

f_noise = 3870;  % calculated noise frequency

%2.signal definitions

%original signal:

%x(t) = sin(2*pi*420*t)

%      - cos(2*pi*190*t)

%      + 0.8*sin(2*pi*250*t)*exp(-12*t)

x = sin(2*pi*420*t) ...

    - cos(2*pi*190*t) ...

    + 0.8*sin(2*pi*250*t).*exp(-2*alpha*t);

%noise signal:

%n(t) = 0.5*cos(2*pi*3870*t)

%      + 0.3*sin(2*pi*6500*t + 2*pi/5)

n = 0.5*cos(2*pi*f_noise*t) ...
```

```

    + 0.3*sin(2*pi*6500*t + 2*pi/5);

%noisy signal
y = x+n;

%3.FFT analysis
N = length(y);

%frequency axis
f = (0:N-1)*(Fs/N);

%positive-frequency portion
h = 1:floor(N/2);

%FFT of clean voice
X = fft(x);

Xm = abs(X)/N;

%FFT of noise
N_fft = fft(n);

Nm = abs(N_fft)/N;

%FFT of noisy signal
Y = fft(y);

Ym = abs(Y)/N;

% 4.butterworth low-pass filter
filter_order = 6;           %based on design not alpha

fc = 1000;                   %cutoff fre in Hz

%normalized cutoff frequency
Wn = fc/(Fs/2);

%design Butterworth LPF

```

```
[b,a] = butter(filter_order, Wn, 'low');
```

```
%apply zero-phase filtering
```

```
yf = filtfilt(b, a, y);
```

```
%FFT of filtered signal
```

```
Yf = fft(yf);
```

```
Yfm = abs(Yf)/N;
```

```
%5.time domain plotting
```

```
%first 0.01 seconds
```

```
tp = t <= 0.01;
```

```
figure;
```

```
subplot(4,1,1)
```

```
plot(t(tp), x(tp), 'b');
```

```
title('x(t) original signal');
```

```
xlabel('time (s)');
```

```
ylabel('amplitude');
```

```
grid on;
```

```
subplot(4,1,2);
```

```
plot(t(tp), n(tp), 'r');
```

```
title('n(t) noise signal');
```

```
xlabel('time (s)');
```

```
ylabel('amplitude');
```

```
grid on;
```

```
subplot(4,1,3);
```

```
plot(t(tp), y(tp), 'm');
```

```

title('y(t) = x(t) + n(t) noisy signal');
xlabel('time (s)');
ylabel('amplitude');
grid on;
subplot(4,1,4);
plot(t(tp), yf(tp), 'g');
title('filtered signal');
xlabel('time (s)');
ylabel('amplitude');
grid on;
sgtitle('time domain analysis');

%6.frequency domain plotting
figure;
subplot(4,1,1);
plot(f(h), 2*Xm(h), 'b');
title('X(f) original voice spectrum');
xlabel('frequency (Hz)');
ylabel('magnitude');
xlim([0 10000]);
grid on;
subplot(4,1,2);
plot(f(h), 2*Nm(h), 'r');
title('N(f) noise spectrum');
xlabel('frequency (Hz)');

```

```

ylabel('magnitude');
xlim([0 10000]);
grid on;
subplot(4,1,3);
plot(f(h), 2*Ym(h), 'm');
title('Y(f)noisy signal spectrum');
xlabel('frequency (Hz)');
ylabel('magnitude');
xlim([0 10000]);
grid on;
subplot(4,1,4);
plot(f(h), 2*Yfm(h), 'g');
title('filtered signal spectrum');
xlabel('frequency (Hz)');
ylabel('magnitude');
xlim([0 10000]);
grid on;
sgtitle('frequency domain analysis');

%7. butterworth filter frequency response
figure;
freqz(b, a, 1024, Fs);
sgtitle('sixth order butterworth lowpass filter response');

%8. SNR calculation
%before filtering

```

```
SNR_past = 10*log10(mean(x.^2)/mean(n.^2));  
  
%leftover error after filtering  
leftover_noise = yf - x;  
  
%after filtering  
SNR_present = 10*log10(mean(x.^2)/mean(leftover_noise.^2));  
  
% improvement  
SNR_cumulation = SNR_present - SNR_past;  
  
fprintf('SNR results\n');  
  
fprintf('SNR before filtering : %.2f dB\n', SNR_past);  
  
fprintf('SNR after filtering : %.2f dB\n', SNR_present);  
  
fprintf('SNR improvement : %.2f dB\n', SNR_cumulation);
```

Output Analysis:

- Time Domain

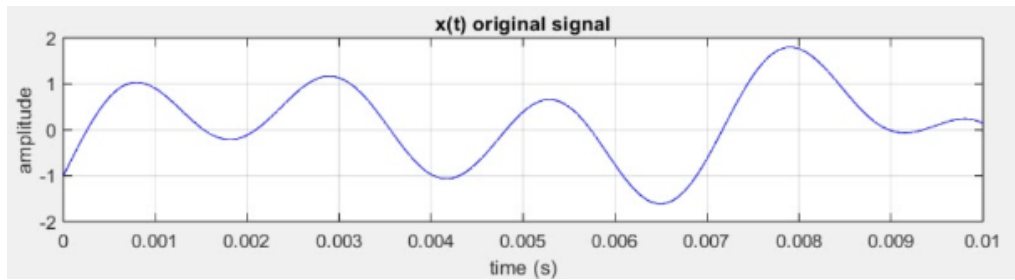


Fig 1: Main Signal $[x(t)]$

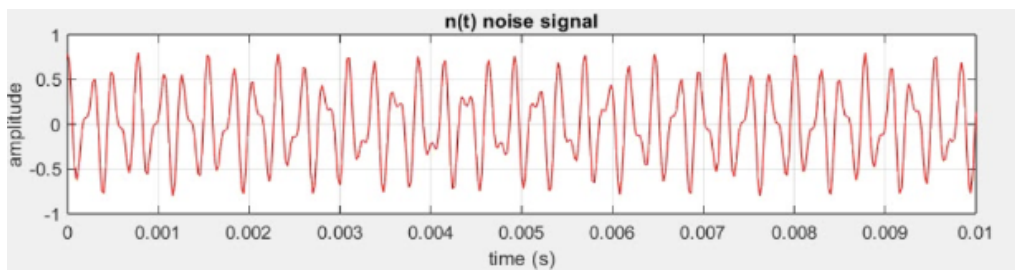


Fig 2: Noise Signal $[n(t)]$

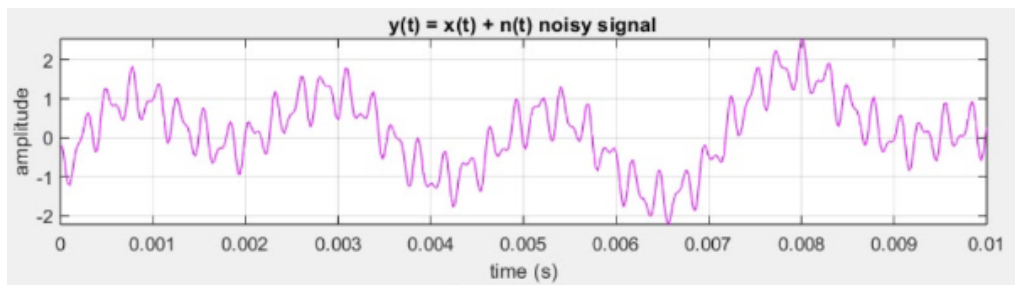


Fig 3: $y(t)=x(t)+n(t)$

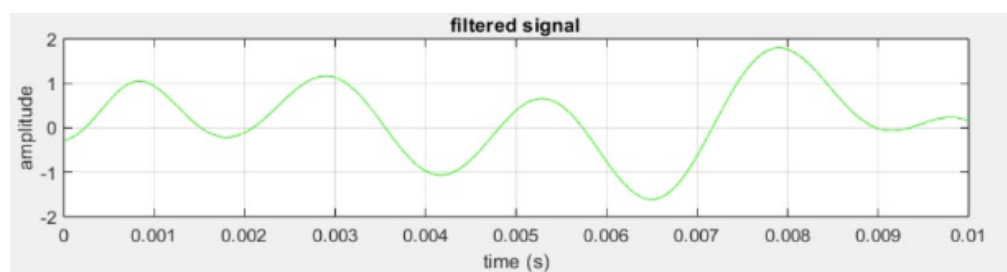


Fig 4: Filtered Signal

- **Frequency Domain**

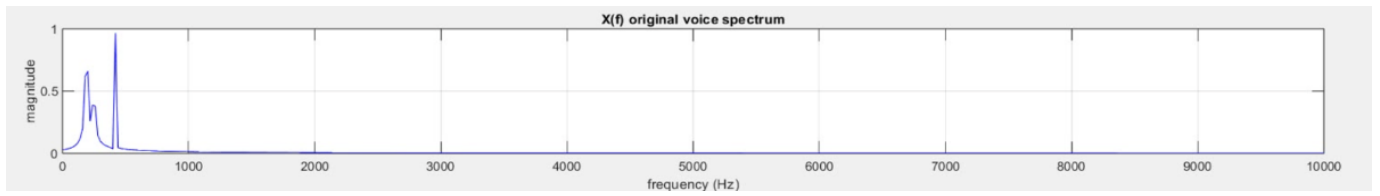


Fig 5: Main Signal $[x(f)]$

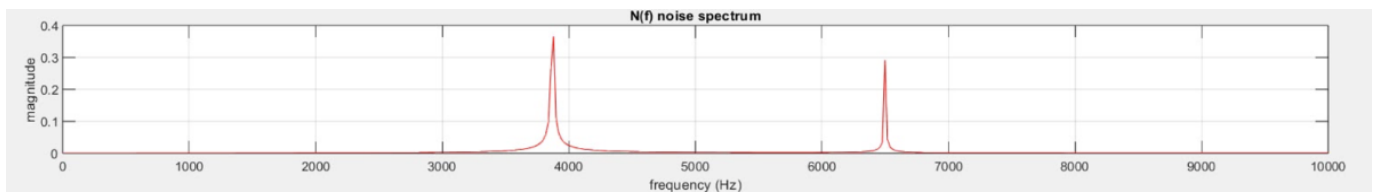


Fig 6: Noise Signal $[n(f)]$

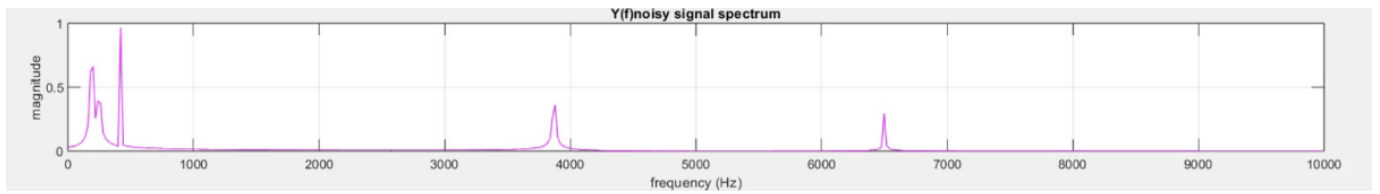


Fig 7: $y(t)=x(f)+n(f)$

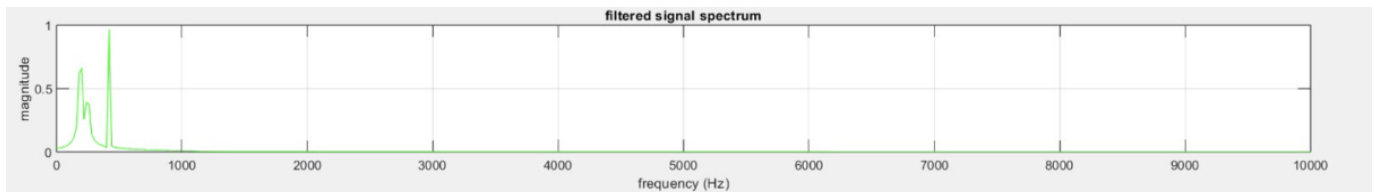


Fig 8: Filtered Signal Spectrum

- **Filter frequency response**

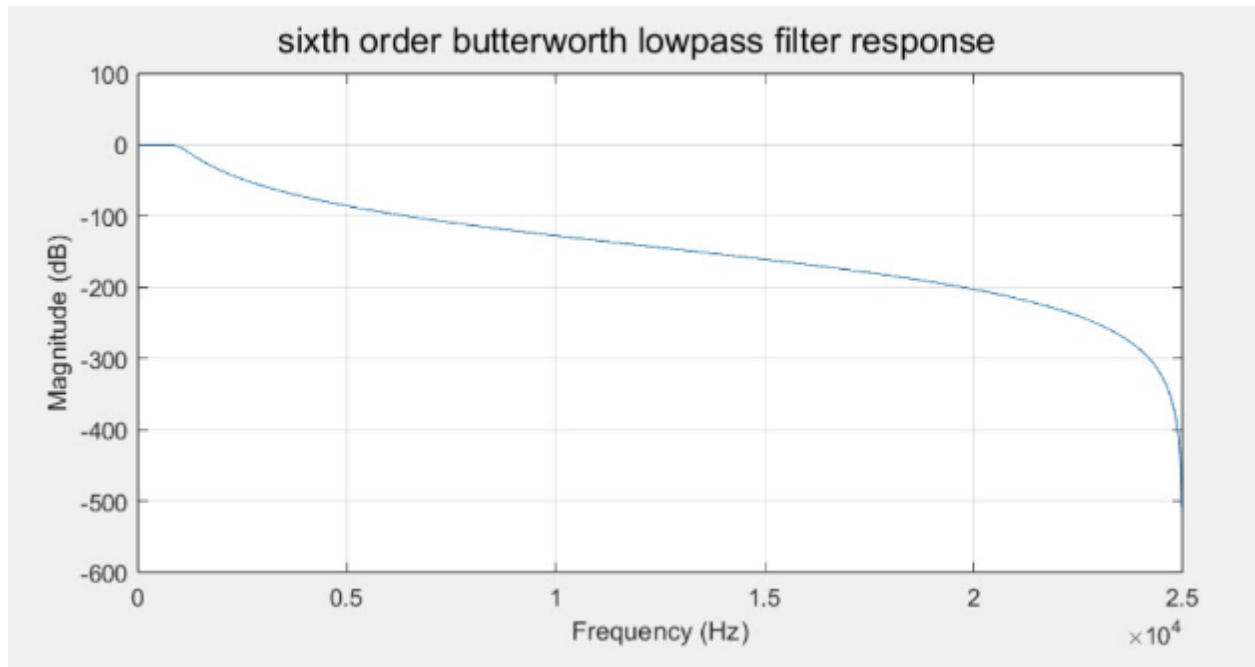


Fig 9: Magnitude Spectrum

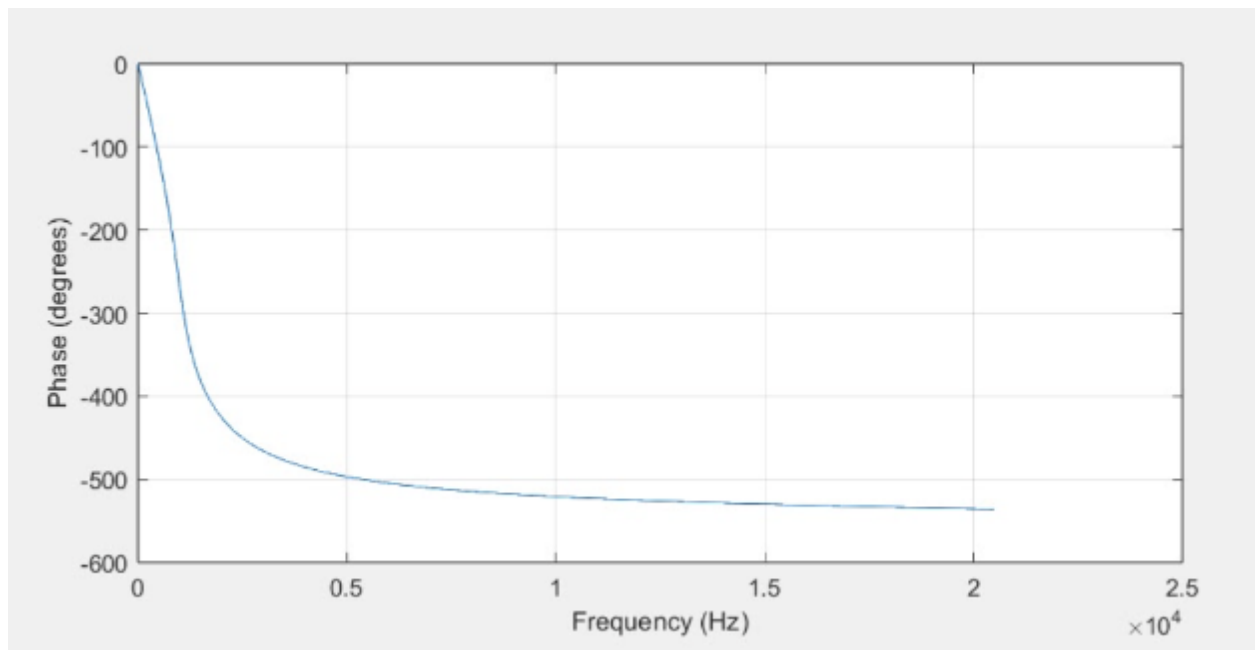


Fig 10:Phase Spectrum

Result Analysis:

- **SNR results**

SNR before filtering : 8.21 dB

SNR after filtering : 28.44 dB

SNR improvement : 20.23 dB

The system was evaluated using Signal-to-Noise Ratio (SNR). Before filtering, the SNR was **8.21 dB**, indicating the signal was significantly corrupted by noise. After applying the Butterworth low-pass filter, the SNR improved to **28.44 dB**, resulting in a gain of **20.23 dB**. This improvement confirms that the filter successfully suppressed the high-frequency noise components while preserving the original voice signal with minimal distortion.

Code Explanation:

1. Signal Definitions

Voice signal, $x(t)$ — consists of three components mimicking a simplified voice signal:

- $\sin(840\pi t)$: a steady sinusoidal component at 420 Hz
- $-\cos(380\pi t)$: a low frequency component at 190 Hz
- $0.8\sin(500\pi t)e^{-12t}$: a damped sinusoidal signal at 250 Hz

Noise signal, $n(t)$ — consists of two high frequency rapid interferences

- $0.5 \cos(7740\pi t)$: calculated noise frequency at 3870 Hz
- $0.3 \sin(13000\pi t + 2\pi/5)$: calculated noise frequency with a phase offset at 6500 Hz

2. Frequency Analysis:

The Fast Fourier Transform is performed on all three signals and filtered to find their frequency content. We plot the spectrum up to 10kHz, confirming where the voice is concentrated, where the noise signal peaks and how the filter removes those noisy signals well.

3. Filter Design & Application:

A 6th order Butterworth low pass filter is used with a cutoff frequency, $f_c = 1000$ Hz, with the MATLAB functions: 'butter ()'.

4. SNR Calculation:

Here, SNR (Signal to Noise Ratio) is computed before filtering to compare the power of $x(t)$ against $n(t)$ and after filtering to compare the power of $x(t)$ with $y(f)-x(t)$.

Performance Analysis:

• Time-Domain Comparison

The time domain analysis represents the efficiency of the low pass filtering process by graphically comparing the original, noise, mixed and processed filtered signals.

Fig 1: Main Signal $[x(t)]$ showcases the clean, low frequency signal with continuous oscillations. Its amplitude varies between -1.6 and +1.8 over 0.01 sec approximately. While *Fig 2: Noise Signal $[n(t)]$* consists of high frequency interference with rapid oscillations with peak amplitude at +0.8 and -0.8.

When the noise signal $n(t)$ is added to produce $y(t)$, the waveform becomes distorted. As seen in *Fig 3: $y(t)=x(t)+n(t)$* the noisy signal presents rapid and irregular fluctuations imposed on the underlying voice signal.

After applying the 6th order Butterworth low pass filter with a cutoff frequency, $f_c = 1000$ Hz, the filtered signal (*Fig 4: Filtered Signal*) the filtered signal resembles the original $x(t)$ voice signal. The rapid high frequency oscillations have been suppressed and the filtered waveform aligns with the original voice signal in shape and amplitude, confirming successful noise removal in the time domain.

• Frequency-Domain Comparison

The frequency domain analysis demonstrates a more precise visual of the noise removal process by examining the spectral of each signal.

As seen in *Fig 5: Main Signal $[x(f)]$* , the spectrum of the original voice signal is concentrated at low frequencies, at 190 Hz, 250 Hz and 420 Hz, which is consistent with the theoretical Fourier Transform. The decaying component of $x(f)$ shows a broad spectral around 250 Hz, as theoretically shown in the *Table: Spectrum Analysis*.

The noise signal *Fig 6: Noise Signal $[n(f)]$* 's frequency spectrum peaks at 3870 Hz and 6500 Hz. In the noisy signal *Fig 7: $y(t)=x(f)+n(f)$* , the spectrum of the two spikes appear alongside the original low frequency noise spectrum. This happens as the noise resides in the high frequency above the voice signals, and it implies that using a low pass filter is an ideal appropriate approach for removing the noise without hindering the voice signal.

After applying the 6th order Butterworth low pass filter with a cutoff frequency, $f_c = 1000$ Hz, the filtered spectrum shown in *Fig 7: $y(t)=x(f)+n(f)$* shows that the noise spikes at 3870 Hz and 6500 Hz have been completely eliminated. The spectral shape of the filtered signal closely matches that of the original voice signal $x(f)$, with energy retained only in the low-frequency region. The smooth roll-off of the Butterworth filter, visible in the magnitude response in *Fig 8: Filtered Signal Spectrum*, ensures that the voice frequency components are preserved without distortion while the noise is attenuated by several hundred decibels beyond the cutoff.